UNIVERSITY OF THE ARMED FORCES ESPE

Computer Science Department

Analysis Of A Web App

Skyscanner Web Platform Analysis

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1 Introduction

Skyscanner is one of the world's leading travel metasearch engines, enabling users to compare millions of flights, hotels, and car rental options from thousands of providers. The Spanish web application represents a sophisticated implementation of modern web development principles, combining complex data visualization with intuitive user interaction patterns.

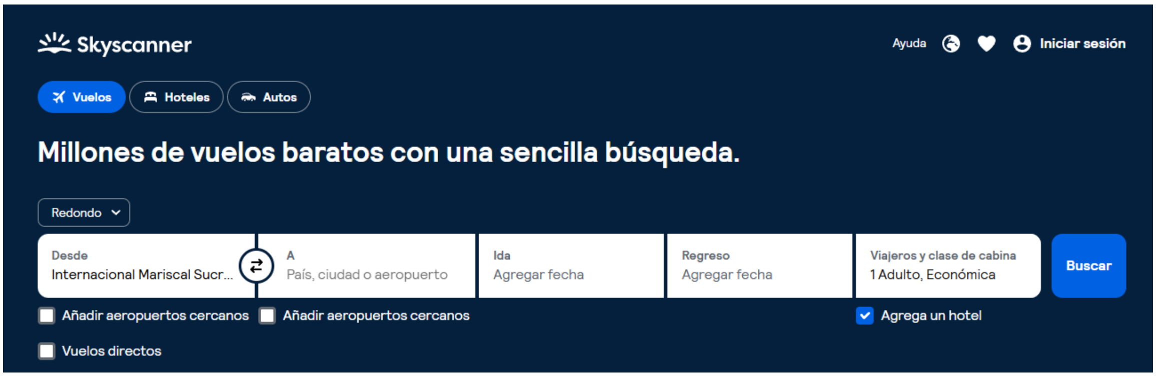

Figure 1: Skyscanner homepage showcasing the primary search interface and main navigation elements

This analysis examines the Skyscanner web platform through the lens of advanced UI/UX design principles, evaluating how the application handles complex data display, implements action patterns, organizes information, facilitates navigation, and captures user input. The study aims to identify both strengths and areas for improvement in the context of contemporary web development standards.

2 Complex Data Display

Skyscanner manages large amounts of travel-related information including flights, hotels, and car rentals. The platform must present this data in a way that is both comprehensive and easily digestible for users making important travel decisions.

2.1 Organizational Models

- **Tabular** → flight results displayed in sortable lists (price, airline, schedule).
- **Hierarchical** → filters for stops, airlines, hotel categories.
- **Geographic** → maps for hotels and destinations.

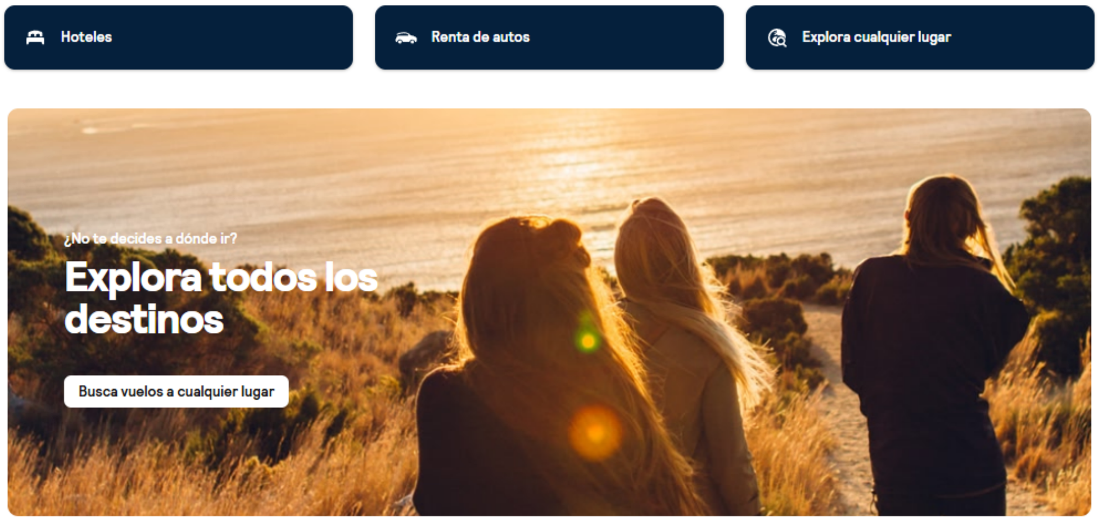

Figure 2: Flight results page demonstrating tabular data organization with sorting capabilities and hierarchical filter system

2.2 Design Elements

- **Preattentive Variables:** color (blue branding, green for best prices), icons (airlines, stars), alignment and size to highlight key data.
- **Techniques:** scroll, filters, sorting, and dynamic updates allow exploration and rearrangement of data.

The interface employs a card-based layout for individual flight options, with each card containing essential information such as departure/arrival times, airline logos, duration, number of stops, and price. The use of whitespace and visual hierarchy ensures that users can quickly scan through multiple options without cognitive overload. Color coding helps distinguish between different price ranges and highlight promotional offers.

3 Actions and Commands

3.1 Interaction Mechanisms

- **Visible mechanisms:** search button, dropdowns for dates and travelers, checkboxes for filters.
- **Invisible mechanisms:** autocomplete for airports/cities, keyboard navigation in date pickers.

3.2 Design Patterns

- **Prominent Done Button** → "Search" is central and emphasized.
- **Smart Menu Items** → contextual filters.
- **Preview** → hotel cards show price and rating before full details.

3.3 Critique

Login requirement for saving searches or price alerts interrupts the natural search flow. While this is a common practice for personalization features, implementing a guest mode with limited save functionality could improve the user experience by reducing friction in the initial exploration phase.

Additionally, the search form could benefit from more contextual help indicators for first-time users, particularly when configuring multi-city trips or flexible date searches. Progressive disclosure techniques could be employed to reveal advanced options without overwhelming casual users.

4 Information Organization

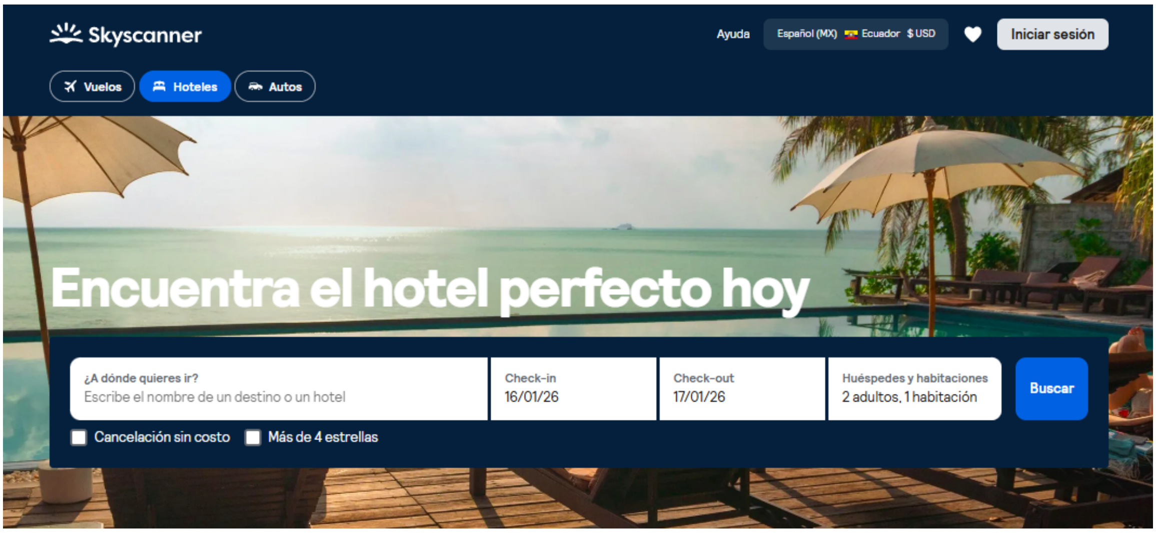

Figure 3: Filter panel demonstrating hierarchical information organization with expandable categories and real-time result updates

The information architecture of Skyscanner follows established patterns for travel booking platforms while introducing innovative elements that enhance usability. The clear separation between different travel services (flights, hotels, car rentals) provides users with a focused experience while maintaining easy access to complementary services.

4.1 Structure

- **Content division:** flights, hotels, and cars separated into tabs.
- **Lists of items:** flights presented as ordered lists; hotels as visual cards.
- **Window structure:** single-page search with expandable panels for filters.

4.2 Patterns Applied

- **Dashboard** → overview of results.
- **Two-panel selector** → list of flights plus expandable details.

5 Navigation

5.1 Orientation Elements

- **Signposts:** logo, highlighted tabs, active filter indicators.
- **"You Are Here" patterns:** tabs and active selections provide orientation, but breadcrumbs are missing.

5.2 Navigational Models

- **Hub-and-spoke** → homepage leads to flights, hotels, cars.
- **Flat navigation** → direct access to main functions.

5.3 Critique

Lack of progress indicators during searches; breadcrumbs could improve orientation in multi-step tasks. The absence of a visible search progress indicator can create uncertainty during the 5-10 second wait time for results to load, particularly for complex multi-leg journeys.

Implementing a skeleton screen or animated progress indicator showing which providers are being searched would provide transparency and reduce perceived wait times. Furthermore, a breadcrumb navigation trail would help users understand their position within the booking flow, especially when comparing multiple options or returning to modify search parameters.

6 Requirements and User Goals

6.1 Actors

- **Casual traveler** → quick searches, "Explore anywhere" option.
- **Planner/power user** → advanced filters, multi-city searches, price alerts.

6.2 Goals

- **Functional goals:** search and filter flights, hotels, cars; save preferences (requires login).
- **Non-functional goals:** lightweight interface, responsive design, mobile accessibility, fast feedback.

7 Getting Input from the User

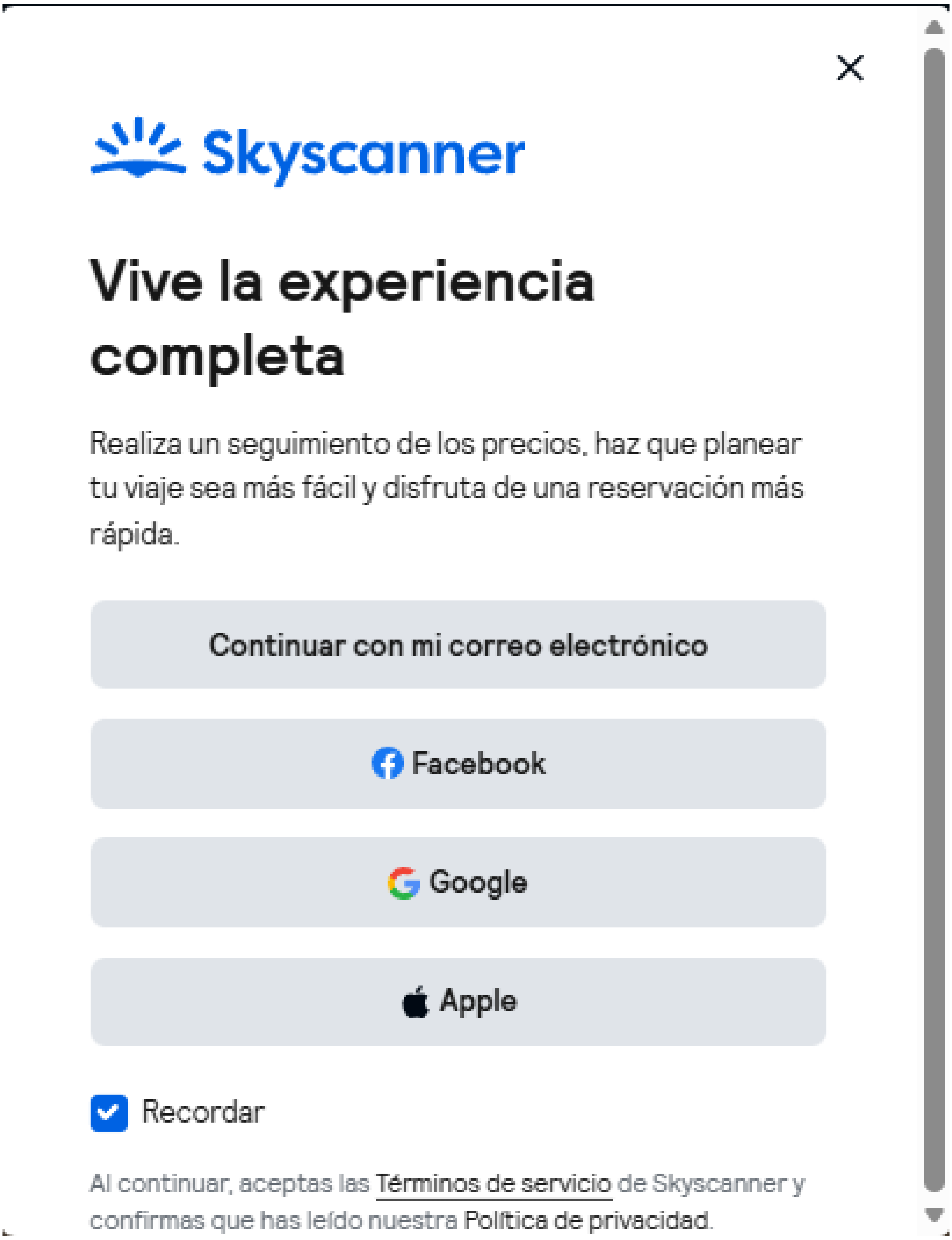

Figure 4: Date picker and passenger selection controls demonstrating forgiving format patterns and smart defaults for streamlined user input

The input mechanisms in Skyscanner are carefully designed to minimize user effort while maximizing flexibility. The platform strikes a balance between providing comprehensive options for power users and maintaining simplicity for casual travelers. Error prevention is built into the system through intelligent validation and contextual constraints.

7.1 Controls Used

- Dropdowns for dates and travelers.
- Autocomplete for airports/cities.
- Checkboxes for multiple filters.
- Calendar picker for dates.

7.2 Patterns

- **Forgiving format** → flexible date selection with visual calendar and month-view options.
- **Autocompletion** → reduces typing errors with intelligent airport/city matching and IATA code recognition.
- **Good defaults** → one adult, economy class preselected based on most common travel scenarios.
- **Input validation** → prevents invalid date ranges and provides real-time feedback.
- **Contextual help** → tooltips and examples guide users through complex inputs.

The date picker deserves special mention for its intelligent design. It prevents selection of past dates, highlights weekends and holidays, and offers flexible search options like "whole month" or "cheapest month" for users with flexible travel dates. The autocomplete functionality for locations goes beyond simple string matching, incorporating fuzzy search, popular destinations, and recently searched locations.

8 Overall Critique

8.1 Strengths

- Clear and centralized input (search box).
- Effective use of autocomplete and dynamic filters.
- Consistent design aligned with travel app conventions.

8.2 Weaknesses

- Heavy reliance on login for advanced features.
- Missing breadcrumbs and progress indicators.
- Perceived latency when loading results; skeleton screens could improve user experience.

9 Conclusion

Skyscanner's Spanish web app is a user-centered travel platform that balances simplicity with advanced functionality. It effectively applies UI design principles such as autocomplete, dashboards, and card layouts, while integrating information organization patterns and action/command mechanisms. However, improvements in navigation feedback and reduction of login friction would enhance usability. Overall, Skyscanner demonstrates a strong application of modern UI/UX methodologies, aligning with the frameworks studied in Advanced Web Development.

The analysis reveals that Skyscanner successfully implements established design patterns while maintaining opportunities for enhancement in areas such as progress indication, breadcrumb navigation, and login flow optimization. These improvements would further elevate the user experience and align the platform with best practices in contemporary web application development.

9.1 Key Takeaways

1. **Data Management:** Skyscanner excels at presenting complex travel data through well-organized tables, filters, and visual hierarchies that facilitate quick decision-making.
2. **User Interaction:** The platform implements sophisticated interaction patterns including autocomplete, dynamic filtering, and contextual previews that reduce cognitive load.
3. **Information Architecture:** Clear separation of services combined with intuitive navigation patterns creates a cohesive user experience across different booking scenarios.
4. **Input Design:** Smart defaults, forgiving formats, and comprehensive validation ensure that users can efficiently input their travel requirements with minimal errors.
5. **Areas for Improvement:** Enhanced progress feedback, reduced authentication friction, and better orientation cues would further optimize the user journey.

This comprehensive analysis demonstrates how Skyscanner applies advanced web development principles to create a robust, scalable, and user-friendly travel search platform. The insights gained from this study can inform future web application design projects, particularly those involving complex data presentation and multi-step user workflows.